



Institute Of Accountancy Arusha

Department Of Information Technology and computer Science

**Bachelor of Computer Science (BCS III) & Bachelor In library and Information Science
(BLIS III) 2025/2026**

Module Name : Computer Security

Module Code: ITU08213

GROUP ASSIGNMENT.

GENERAL INSTRUCTIONS:

1. This is a group assignment which carries a total of **20%** of your coursework Marks.
2. Each group should have 15 members only and not more. **(expecting to have 8 groups, each group should work on one question as the number reflects to your group numbers)**
3. The submission date for this work will be on 25th May 2026.
4. **Prepare a 10 slides PowerPoint for presentation and there will be a group presentation. (dates will be shared after submission of the report)**
5. **You are required to prepare a compressive Notes (a 3000+ words word document) on the topic provided by analyzing with examples, the following issues;**

1. Explain the benefits of digital certificates and digital signatures in securing web mail application systems.
2. Discuss the concept of encryption in Information security. Provide a thorough analysis on the best encryption algorithm used in securing communication over the internet.
3. Discuss the Application of protocol stack to secure web applications.
4. Discuss how blockchain technology ensures data integrity and security through cryptographic principles.
5. Discuss the problems caused by buffer overflow attacks in computer security.
6. Explain cryptography as related to computer security and suggest possible solutions to Overcome attacks caused by different cryptanalytic.
7. Discuss the application of handshake protocol in securing web applications.
8. Discuss the problems posed by cybercrime in network security.